NETPILOT PROJECT REPORT

Office / Small-Business Network

Design, configuration, audit and security hardening in Cisco Packet Tracer

Item	Detail
Project owner	**Michel**
Network name	Michel's Office Network
Prepared by	NetPilot
Tool used	Cisco Packet Tracer (lab file Small-Business-Network.pkt)
Date	October 12, 2026
Version	2 - adds the audit, the fixes applied and the device passwords

Figure placeholders. Every dashed box marked **INSERT FIGURE n HERE** shows the exact spot for a Packet Tracer screenshot. Paste the screenshot over the box and keep the caption below it. The List of Figures at the end matches these numbers.

Confidential. Section 10 lists the lab passwords. Keep this report private, and use new passwords on any real network.

1. Introduction

This report describes the design, configuration, testing and audit of a small-business office network built for Michel. The network connects eight computers in four departments to one router and one switch, and gives them access to the Internet. Each department sits in its own VLAN (virtual LAN) and its own IP subnet, and the router lets the departments exchange messages with each other.

After the build, the lab was audited by checking every device configuration against the actual cabling. The audit found four items. One blocked connectivity and was fixed, two were security improvements and were applied, and one was informational. Sections 9 and 10 describe each step.

- **VLANs** to separate departments.
- **802.1Q trunking** to carry all VLANs over one cable between the switch and the router.
- **Inter-VLAN routing** (router-on-a-stick) so departments can communicate.
- **DHCP** pools on R1, available for automatic addressing.
- **NAT (overload)** so the user departments share one public address on the Internet.
- **Device passwords** on the ISP router, R1 and SW1.

2. Objectives

1. Build the topology in Packet Tracer with the correct devices and cables.
2. Create one VLAN and subnet per department and explain why each is separate.
3. Give every device an IP address.
4. Make every computer able to send messages to every other computer.
5. Give the user departments access to the Internet through NAT.
6. Protect the network devices with passwords.
7. Verify the result with tests and record every problem found and fixed.

3. Network Design

3.1 Why each segment is separate

Segment	Reason for separation
VLAN 10 Executive	Keeps Michel's and management's files and email in a protected group.
VLAN 20 Sales	Handles outside customers and a lot of email, so it is the most exposed to phishing. Separating it limits the damage.
VLAN 30 Accounting	Holds payroll and financial records, so access should be easy to restrict.
VLAN 40 IT	Administrators need tools and access that normal staff should not have.
VLAN 99 Network management	Keeps switch and router admin addresses away from user computers and off the Internet path.

Table 1: Network segments and the reason for each.

Separating the network also gives smaller broadcast domains (faster network), easier troubleshooting, and room to grow.

3.2 Topology

The ISP router connects to R1 over the WAN link (203.0.113.0/30). R1 connects to the switch SW1 through one 802.1Q trunk carrying VLANs 10, 20, 30, 40 and 99. The eight computers connect to SW1 access ports. The ISP router has a loopback address, 8.8.8.8, that stands in for the Internet.

INSERT FIGURE 1 HERE

Screenshot of the complete Packet Tracer workspace showing ISP, R1, SW1 and all eight computers with their names visible.

Figure 1: Network topology of Michel's Office Network in Packet Tracer.

3.3 How a message travels

Inside a VLAN, the switch delivers the message directly from one port to the other using MAC addresses. Between VLANs, the message goes up the trunk to R1, R1 routes it to the destination subnet, and it returns down the same trunk to the correct port.

4. Devices Used

Qty	Device	Packet Tracer model	Name
1	Router (Internet side, for testing)	Cisco 2911	ISP
1	Router (office)	Cisco 2911	R1
1	Switch	Cisco 2960-24TT	SW1
8	Computers	PC-PT (generic PC)	Michel-PC, Exec-Laptop, Sales-PC1, Sales-PC2, Acct-PC1, Acct-PC2, IT-PC1, IT-PC2

Table 2: Devices used in the project.

The lab file also contains a Power Distribution Device. It has no network links, which is expected.

INSERT FIGURE 2 HERE

Screenshot of the Packet Tracer device palette or the placed devices, showing the router, switch and PC models used.

Figure 2: Devices placed in the workspace.

5. Cabling

The computers are cabled to SW1 ports Fa0/1 to Fa0/8. Table 3 shows the cabling as built. The original plan used ports Fa0/5, 6, 9, 10, 13 and 14 for the later departments, and the switch ports were changed to match the real cabling (see section 9.1).

From	Port	To	Port	Cable
ISP	Gi0/0	R1	Gi0/1	Copper crossover
R1	Gi0/0	SW1	Gi0/1	Copper straight-through
SW1	Fa0/1	Michel-PC	FastEthernet0	Copper straight-through
SW1	Fa0/2	Exec-Laptop	FastEthernet0	Copper straight-through
SW1	Fa0/3	Sales-PC1	FastEthernet0	Copper straight-through
SW1	Fa0/4	Sales-PC2	FastEthernet0	Copper straight-through
SW1	Fa0/5	Acct-PC1	FastEthernet0	Copper straight-through
SW1	Fa0/6	Acct-PC2	FastEthernet0	Copper straight-through
SW1	Fa0/7	IT-PC1	FastEthernet0	Copper straight-through
SW1	Fa0/8	IT-PC2	FastEthernet0	Copper straight-through

Table 3: Cabling as built (10 connections: 9 straight-through, 1 crossover).

INSERT FIGURE 3 HERE

Close-up screenshot of the switch SW1 and its cables with all link lights green.

Figure 3: Cabling between SW1, R1 and the computers with all links up.

6. IP Addressing Plan

Each department has one /24 subnet (mask 255.255.255.0). The router is always address .1. Addresses .1 to .50 are reserved for infrastructure and fixed computers, and the DHCP pools hand out .51 to .254.

VLAN	Name	Network	Gateway (R1)
10	Executive	192.168.10.0/24	192.168.10.1
20	Sales	192.168.20.0/24	192.168.20.1
30	Accounting	192.168.30.0/24	192.168.30.1
40	IT	192.168.40.0/24	192.168.40.1
99	Network management	192.168.99.0/24	192.168.99.1

Table 4: VLANs and subnets.

Device	Interface	IP address	Mask	Gateway
ISP	Gi0/0	203.0.113.1	255.255.255.252	-
ISP	Loopback0 (test)	8.8.8.8	255.255.255.255	-
R1	Gi0/1 (WAN)	203.0.113.2	255.255.255.252	203.0.113.1 (default route)
R1	Gi0/0.10 / .20 / .30 / .40 / .99	192.168.10.1 / 20.1 / 30.1 / 40.1 / 99.1	255.255.255.0	-
SW1	Vlan99	192.168.99.2	255.255.255.0	192.168.99.1
Michel-PC	FastEthernet0	192.168.10.11	255.255.255.0	192.168.10.1
Exec-Laptop	FastEthernet0	192.168.10.12	255.255.255.0	192.168.10.1
Sales-PC1	FastEthernet0	192.168.20.11	255.255.255.0	192.168.20.1
Sales-PC2	FastEthernet0	192.168.20.12	255.255.255.0	192.168.20.1
Acct-PC1	FastEthernet0	192.168.30.11	255.255.255.0	192.168.30.1
Acct-PC2	FastEthernet0	192.168.30.12	255.255.255.0	192.168.30.1
IT-PC1	FastEthernet0	192.168.40.11	255.255.255.0	192.168.40.1
IT-PC2	FastEthernet0	192.168.40.12	255.255.255.0	192.168.40.1

Table 5: IP addresses of every device. All computers use DNS 8.8.8.8.

7. Configuration

7.1 ISP router

The ISP router was given the WAN address 203.0.113.1/30 on Gi0/0 and a loopback interface with address 8.8.8.8 so the office has something to reach on the "Internet".

INSERT FIGURE 4 HERE

Screenshot of the ISP CLI showing the output of show ip interface brief (Gi0/0 = 203.0.113.1, Loopback0 = 8.8.8.8).

Figure 4: ISP interface configuration.

7.2 Office router R1

R1 was configured with:

- One sub-interface per VLAN on Gi0/0 (Gi0/0.10, .20, .30, .40, .99), each using encapsulation dot1Q and the .1 gateway address.
- The WAN interface Gi0/1 with 203.0.113.2/30 and a default route to 203.0.113.1.
- NAT overload using access list 1, which permits only the four user subnets 192.168.10.0, 20.0, 30.0 and 40.0 (see section 9.3).
- Four DHCP pools (Executive, Sales, Accounting, IT) with addresses .1 to .50 excluded. The computers currently use static addresses, so the pools are unused (see section 9.4).

INSERT FIGURE 5 HERE

Screenshot of R1 show ip interface brief showing Gi0/1 = 203.0.113.2 and all five sub-interfaces up/up.

Figure 5: R1 interface status.

INSERT FIGURE 6 HERE

Screenshot of R1 show ip route (connected networks and the default route) or of the DHCP and NAT configuration.

Figure 6: R1 routing table and NAT / DHCP configuration.

7.3 Switch SW1

SW1 was configured with five VLANs (10, 20, 30, 40, 99), access ports for each department, and a trunk on Gi0/1 allowing VLANs 10, 20, 30, 40 and 99. The management address 192.168.99.2 was set on interface Vlan99 with default gateway 192.168.99.1. Table 6 shows the final port assignment after the fix in section 9.1.

VLAN	Name	Ports in use (final)	Other ports
10	Executive	Fa0/1 - Fa0/2	-
20	Sales	Fa0/3 - Fa0/4	-
30	Accounting	Fa0/5 - Fa0/6	Fa0/9 - Fa0/12 set to VLAN 30, unused
40	IT	Fa0/7 - Fa0/8	Fa0/13 - Fa0/16 set to VLAN 40, unused
99	NetMgmt	none	Management interface Vlan99 only

Table 6: VLAN to switch port assignment (final).

INSERT FIGURE 7 HERE

Screenshot of SW1 show vlan brief.

Figure 7: VLANs and their ports on SW1.

INSERT FIGURE 8 HERE

Screenshot of SW1 show interfaces trunk showing Gi0/1 trunking with VLANs 10, 20, 30, 40, 99.

Figure 8: Trunk link between SW1 and R1.

7.4 Computers

Each computer was given a static IP address, mask, gateway and DNS server from Table 5 using Desktop > IP Configuration.

INSERT FIGURE 9 HERE

Screenshot of the IP Configuration window of Michel-PC (192.168.10.11, 255.255.255.0, 192.168.10.1, 8.8.8.8).

Figure 9: IP configuration of Michel-PC (the owner's computer).

8. Testing and Results

8.1 Early results (before the fixes)

Test	From	To	Result
ISP and R1 communicate	R1	ISP 203.0.113.1	Success
Own gateway	Michel-PC	192.168.10.1	4 of 4 replies
Internet through NAT	Michel-PC	8.8.8.8	3 of 3 replies
Switch to router	SW1	192.168.99.1	4 of 5 replies (first packet lost while learning the address, which is normal)
Switch to Sales gateway	SW1	192.168.20.1	5 of 5 replies
Sales gateway	Sales-PC1	192.168.20.1	Failed (100% loss) - cause in section 9.1

Table 7: Test results recorded before the VLAN port fix.

INSERT FIGURE 10 HERE

Screenshot of the Michel-PC Command Prompt showing the successful pings to 192.168.10.1 and 8.8.8.8.

Figure 10: Michel-PC reaching its gateway and the Internet.

8.2 Results after the fixes

After the switch ports were corrected, each department was tested from a computer's Command Prompt: its own gateway, the other computer in the same VLAN, and the Internet address 8.8.8.8.

Department	Own gateway	Other PC in same VLAN	Internet 8.8.8.8
Executive (VLAN 10)	4 of 4 replies	Not recorded	3 of 3 replies
Sales (VLAN 20)	Success, 0% loss	Success, 0% loss	Success, 0% loss
Accounting (VLAN 30)	Success, 0% loss	Success, 0% loss	Success, 0% loss
IT (VLAN 40)	Success, 0% loss	Success, 0% loss	Success, 0% loss

Table 8: Test results after the VLAN port fix.

8.3 Communication between departments

To confirm that every computer can send messages to every other computer, run the ping tests below and record the result in the last column.

From	To	Path	Result
Michel-PC	Exec-Laptop (192.168.10.12)	Same VLAN, switch only	
Michel-PC	Sales-PC1 (192.168.20.11)	Via R1	
Michel-PC	Acct-PC1 (192.168.30.11)	Via R1	
Michel-PC	IT-PC1 (192.168.40.11)	Via R1	
Sales-PC1	Acct-PC2 (192.168.30.12)	Via R1	
IT-PC1	Every other computer	Via R1	
Any computer	SW1 (192.168.99.2)	Via R1 to VLAN 99	

Table 9: Inter-department communication test plan and results.

INSERT FIGURE 11 HERE

Screenshot of a Command Prompt showing successful pings from one department to another (for example Sales-PC1 to Acct-PC1).

Figure 11: Successful communication between two different VLANs through R1.

INSERT FIGURE 12 HERE

Screenshot of Packet Tracer Simulation mode showing a Simple PDU (envelope) message sent from Michel-PC to another computer with the status Successful.

Figure 12: A message sent directly between two computers in Packet Tracer.

9. Audit Findings and Fixes

Each device configuration was cross-checked against the physical cabling. A configuration can look correct on one device and still be wrong relative to how everything is wired together, and this check found the main defect. Four items were found. They are ordered by whether they broke connectivity (must fix) or were security gaps (should fix).

#	Issue	Severity	Status
1	SW1 access ports did not match the VLAN implied by each computer's static IP	Blocking: broke connectivity	Fixed
2	No enable, VTY or console passwords on the three devices	Security	Fixed
3	NAT access list also translated the management subnet 192.168.99.0/24	Security	Fixed
4	DHCP pools on R1 unused because all computers use static addresses	Informational	Not a defect, no change

Table 10: Audit findings.

9.1 Issue 1: VLAN and port mismatch (blocking)

What was wrong. SW1's access ports were set for a generic four-ports-per-department layout (ports 1 to 4, 5 to 8, 9 to 12 and 13 to 16). The computers were actually cabled to ports 1 to 8, in a different order, so six of the eight were in the wrong VLAN.

Port	Configured VLAN	Computer connected	Its real subnet	Match?
Fa0/1	10	Michel-PC	192.168.10.x	Correct
Fa0/2	10	Exec-Laptop	192.168.10.x	Correct
Fa0/3	10	Sales-PC1	192.168.20.x	Wrong
Fa0/4	10	Sales-PC2	192.168.20.x	Wrong
Fa0/5	20	Acct-PC1	192.168.30.x	Wrong
Fa0/6	20	Acct-PC2	192.168.30.x	Wrong
Fa0/7	20	IT-PC1	192.168.40.x	Wrong
Fa0/8	20	IT-PC2	192.168.40.x	Wrong

Table 11: Port assignments before the fix.

Effect. Traffic from those six computers was tagged into the wrong VLAN on the trunk to R1. A computer asking for its own gateway, for example Sales-PC1 asking for 192.168.20.1, reached a VLAN where that address does not exist, so Sales, Accounting and IT had no connectivity.

INSERT FIGURE 13 HERE

Screenshot of SW1 show interfaces status showing the problem (computers on Fa0/1 to Fa0/8, Fa0/9 to Fa0/16 not connected, VLANs not matching).

Figure 13: SW1 port status before the fix.

Fix, step by step. The switch was changed to match the cables, one port at a time.

Step 1. Open SW1 and enter privileged mode, then configuration mode.

Step 2. Move each wrongly assigned port to the VLAN of the computer on it.

Step 3. Leave configuration mode, then save.

```
enable
configure terminal
interface FastEthernet0/3
 switchport access vlan 20
interface FastEthernet0/4
 switchport access vlan 20
interface FastEthernet0/5
 switchport access vlan 30
interface FastEthernet0/6
 switchport access vlan 30
interface FastEthernet0/7
 switchport access vlan 40
interface FastEthernet0/8
 switchport access vlan 40
end
copy running-config startup-config
```

Step 4. Verify with show interfaces status, then ping from each department (results in Table 8).

INSERT FIGURE 14 HERE

Screenshot of SW1 show interfaces status after the fix, with each computer on its planned port and VLAN.

Figure 14: SW1 port status after the fix.

9.2 Issue 2: no device passwords (security)

What was wrong. None of R1, the ISP router or SW1 had an enable password, a console password or a working remote login. The remote lines had login set with no password behind them, and anyone opening a device's CLI landed straight in privileged mode.

Fix, step by step. The same setup was applied to all three devices, with different passwords on each. The exact values are in section 10.

Command	What it does
service password-encryption	Hides plain-text line passwords in the configuration (weak, type 7).
enable secret ...	Protects privileged mode with a hashed password (type 5).
username admin secret ...	Creates a local administrator account with a hashed password.
line console 0 / password ... / login	Asks for a password on the console cable.
line vty 0 4 / login local	Remote logins must use the local admin account. SW1 also has line vty 5 15.

Table 12: Security commands and their purpose.

Result. Each device now shows an `enable secret 5` line, a `username admin secret 5` line, a console password with login, and `login local` on every remote line. Each device uses different passwords.

9.3 Issue 3: NAT list covered the management VLAN (security)

What was wrong. Access list 1 permitted the whole range 192.168.0.0 to 192.168.255.255, which includes VLAN 99. Management traffic was therefore allowed to share the Internet address translation with user traffic, which is not wanted for a management network.

Fix, step by step. On R1, remove the wide entry and permit only the four user subnets. The NAT statement already points at access list 1, so it needed no change.

```
enable
configure terminal
no access-list 1 permit 192.168.0.0 0.0.255.255
access-list 1 permit 192.168.10.0 0.0.0.255
access-list 1 permit 192.168.20.0 0.0.0.255
access-list 1 permit 192.168.30.0 0.0.0.255
access-list 1 permit 192.168.40.0 0.0.0.255
end
copy running-config startup-config
```

Result. Access list 1 now has exactly four lines. The user departments still reach the Internet, and VLAN 99 stays internal.

INSERT FIGURE 15 HERE

Screenshot of R1 show access-lists showing access list 1 with the four permitted subnets (192.168.10.0, 20.0, 30.0 and 40.0).

Figure 15: NAT access list on R1 after the fix.

9.4 Issue 4: unused DHCP pools (informational)

R1 has valid DHCP pools for the four user VLANs (EXECUTIVE, SALES, ACCOUNTING, IT), each with the correct network, default router and DNS server, and with the first 50 addresses excluded. Every computer uses a static address, so the pools are unused. This is not a misconfiguration: the excluded range lets static and automatic addressing coexist. No change was made, and the pools stay available if a later version of the lab shows DHCP.

10. Device Access Credentials

These are the passwords set on the three devices. To log in, open the device's CLI, type `enable` and enter the enable secret. For remote logins, use the username admin with the admin password.

Device	Enable secret	Admin username	Admin password	Console password
ISP	ISP-0fficePass	admin	Admin-ISP-0fficePass	C0nsole-ISP!
R1	R1-0fficePass	admin	Admin-R1-0fficePass	C0nsole-R1!
SW1	SW1-0fficePass	admin	Admin-SW1-0fficePass	C0nsole-SW1

Table 13: Passwords set on each device.

Read this carefully. The character after the hyphen in "0fficePass" is the **digit zero**, not the letter O. In "C0nsole" the second character is also the digit zero. Passwords are case-sensitive. The SW1 console password has no exclamation mark; the ISP and R1 console passwords do.

Lab use only. These passwords are written in this report for the lab. On a real network, choose new passwords and store them in a password manager.

Exact commands entered on each device

ISP

```
enable
configure terminal
service password-encryption
enable secret ISP-0fficePass
username admin secret Admin-ISP-0fficePass
line console 0
```

```
 password C0nsole-ISP!
 login
line vty 0 4
 login local
end
copy running-config startup-config
```

R1

```
enable
configure terminal
service password-encryption
enable secret R1-0fficePass
username admin secret Admin-R1-0fficePass
line console 0
 password C0nsole-R1!
 login
line vty 0 4
 login local
end
copy running-config startup-config
```

SW1 (the switch has 16 remote lines, so lines 5 to 15 are secured too)

```
enable
configure terminal
service password-encryption
enable secret SW1-0fficePass
username admin secret Admin-SW1-0fficePass
line console 0
 password C0nsole-SW1
 login
line vty 0 4
 login local
line vty 5 15
 login local
end
copy running-config startup-config
```

INSERT FIGURE 16 HERE

Screenshot of R1 (or SW1) show running-config | include username|enable secret, showing the hashed enable secret and admin username lines.

Figure 16: Hashed passwords in the device configuration.

INSERT FIGURE 17 HERE

Screenshot of the SW1 CLI asking for a password after typing enable, then reaching the SW1# prompt.

Figure 17: Password prompt on SW1.

11. Final State

Area	Before	After
Sales, Accounting and IT connectivity	Broken: wrong VLAN on six of eight ports	Working: gateway, same-VLAN and Internet pings succeed
Device administration	Open: no enable, console or remote password	Protected: hashed enable secret, local login for remote access, console password, different on each device
NAT scope	Included the management subnet	Limited to the four user VLANs
DHCP pools	Configured, unused	Unchanged, available for future use

Table 14: Network state before and after the audit.

The lab now matches its design: four separate department VLANs routed by R1, translated to a simulated Internet through the ISP loopback 8.8.8.8, with basic administrative access control and the management network kept off the NAT path.

12. Conclusion

The project built a segmented office network for Michel with separate VLANs for Executive, Sales, Accounting and IT, a management VLAN for the network devices, inter-VLAN routing on R1, and NAT for Internet access. The audit found that six computers were in the wrong VLAN because the switch ports did not match the cabling. Fixing the ports restored Sales, Accounting and IT. Passwords were added to all three devices and the NAT list was narrowed to the user VLANs. The remaining step is to run the cross-department tests in Table 9 and record the results.

13. Recommendations and Next Steps

- Run the cross-department pings in Table 9 and the Simple PDU message test, and add the screenshots (figures 11 and 12).
- Use SSH instead of Telnet on the remote lines (transport input ssh, plus an RSA key). Telnet is still allowed today.
- Add access control lists on R1 to restrict sensitive departments, for example blocking Sales from Accounting and the management VLAN.
- Shut down unused switch ports and move them to an unused VLAN.

- Switch one computer per VLAN from static to DHCP to demonstrate the existing pools.
- Add a guest Wi-Fi VLAN, or a second switch or access point, if wireless access is needed later.
- Save the configuration (copy running-config startup-config) on every device after each change.

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Sign-off

Project owner	Michel	Signature	
Prepared by	NetPilot	Date	October 12, 2026